

Intercalation-Modulated Electronic Regulation of Nitrogen Reduction on Doped MoS₂

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Abstract

Electrochemical nitrogen reduction reaction (NRR) is a promising alternative to the Haber–Bosch process but is fundamentally limited by sluggish kinetics, poor selectivity against the hydrogen evolution reaction, and the difficulty of balancing N₂ activation with protonation energetics. Here, we report a reconstruction-aware computational study of NRR on intercalated and doped MoS₂, integrating high-throughput density functional theory calculations with electronic-structure analysis and explainable machine learning. Screening 784 dopant–intercalant combinations reveals that NRR activity is primarily dictated by the doped atom, with Mo and W emerging as optimal active centers, while intercalation serves as an electronic regulator that modulates selectivity by redistributing Mo-projected d-states near the Fermi level, tuning the balance between σ -donation and π -back-donation at the active site without perturbing surface coordination geometry. Mechanistic analysis identifies NH # NH₂ as the dominant potential-determining step for most active and selective systems, revising the long-standing emphasis on early or late hydrogenation steps. We further establish a transferable descriptor space defined by N₂ adsorption free energy, N₂ charge state, and N–N bond length that robustly captures both activity and selectivity across the full chemical space. Grand-canonical DFT validation identifies Hf-intercalated Mo–MoS₂ as a priority experimental target. These results provide generalizable design principles for selective NRR catalysts on layered materials.

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Abstract:

Electrochemical nitrogen reduction reaction (NRR) is a promising alternative to the Haber–Bosch process but is fundamentally limited by sluggish kinetics, poor selectivity against the hydrogen evolution reaction, and the difficulty of balancing N₂ activation with protonation energetics. Here, we report a reconstruction-aware computational study of NRR on intercalated and doped MoS₂, integrating high-throughput density functional theory calculations with electronic-structure analysis and explainable machine learning. Screening 784 dopant–intercalant combinations reveals that NRR activity is primarily dictated by the doped atom, with Mo and W emerging as optimal active centers, while intercalation serves as an electronic regulator that modulates selectivity by redistributing Mo-projected d-states near the Fermi level, tuning the balance between σ -donation and π -back-donation at the active site without perturbing surface coordination geometry. Mechanistic analysis identifies NH $\rightarrow$ NH₂ as the dominant potential-determining step for most active and selective systems, revising the long-standing emphasis on early or late hydrogenation steps. We further establish a transferable descriptor space defined by N₂ adsorption free energy, N₂ charge state, and N–N bond length that robustly captures both activity and selectivity across the full chemical space. Grand-canonical DFT validation identifies Hf-intercalated Mo–MoS₂ as a priority experimental target. These results provide generalizable design principles for selective NRR catalysts on layered materials.

1. Introduction

Electrochemical nitrogen reduction reaction (NRR) offers a sustainable pathway for ammonia synthesis under ambient conditions, providing an attractive alternative to the energy-intensive Haber–Bosch process that operates at 400–500 °C and 15–25 MPa with substantial carbon emissions.^{1–4} However, the NRR remains fundamentally constrained by the exceptional thermodynamic stability of the N $\equiv$ N triple bond (dissociation energy $\sim$ 941 kJ mol^{–1}), which imposes severe kinetic barriers on N₂ activation at or near room temperature.⁵ In aqueous electrolytes, the challenge is compounded by low N₂ solubility and the dominance of the competing hydrogen evolution reaction (HER), which shares the same potential window and drastically reduces Faradaic efficiency.^{6,7} Consequently, achieving a favorable balance among catalytic activity, selectivity against HER, and

long-term structural stability remains a central bottleneck, and meeting all three criteria simultaneously within a single catalyst material demands rational design strategies that go beyond trial-and-error optimization.^{8–10}

Layered MoS₂ has emerged as a versatile platform for NRR electrocatalysis, owing in large part to its structural analogy with the MoFe cofactor of biological nitrogenase, its tunable electronic structure, rich defect chemistry, and broad compatibility with heteroatom engineering.^{11–16} In particular, sulfur vacancies on the MoS₂ basal plane have been shown to expose undercoordinated Mo sites that enhance N₂ chemisorption and lower the activation barrier for subsequent protonation steps.^{17–19} Building on this defect-mediated activity, substitutional doping with transition metals (e.g., Fe, Co, Ni) and non-metals (e.g., P, B) has proven effective at further modulating N₂ adsorption and reaction energetics by altering local charge distribution, d-band characteristics, and even inducing beneficial 2H-to-1T phase transitions that increase electrical conductivity.^{20–26} Meanwhile, interlayer intercalation of guest species into the van der Waals gap provides a complementary route to regulate the electronic environment through charge transfer, lattice expansion, and phase engineering, all without directly perturbing the surface coordination geometry.^{27–29} Despite growing interest in both strategies, most of the existing literature treats doping and intercalation as independent approaches, and systematic studies that probe how their cooperative interplay governs N₂ adsorption strength, protonation selectivity, and catalyst reconstruction under operating conditions remain limited.³⁰ This distinction is important because synergistic electronic effects between dopants and intercalants, such as simultaneous tuning of the d-band center and interlayer charge transfer, may expand the accessible performance landscape. However, a systematic understanding of how specific dopant–intercalant combinations jointly influence catalytic descriptors remains underdeveloped.

Here, we establish an integrated catalyst design and screening framework for intercalated and doped MoS₂ (ID@MoS₂), as schematically illustrated in **Figure 1**. By constructing an expanded combinatorial design space that pairs substitutes with intercalated atoms in the presence of sulfur vacancies, we first perform reconstruction-aware stability screening to identify physically meaningful catalyst configurations and eliminate those prone to structural collapse. Subsequent energetic evaluation of key NRR elementary steps including N₂ adsorption, first protonation, and selectivity-determining competition with HER, enables systematic elimination of low-performance systems and reveals clear activity–selectivity trends across the chemical space. A key finding is that doping and intercalation play distinct yet complementary mechanistic roles: dopants primarily tune local binding geometry and d-orbital interactions at the active site, while intercalants modulate interlayer charge transfer, thereby altering the electronic structure of the active site. Building on these mechanistic insights, we identify a compact and physically interpretable descriptor space comprising three readily computed N₂-derived quantities: N–N bond length, N₂ charge state, and N₂ adsorption free energy. Together, these parameters capture bond activation, charge transfer, and thermodynamic binding,^{31,32} preserving mechanistic transparency and enabling clear interpretation of why specific dopant–intercalant combinations enhance or suppress catalytic activity and selectivity.³³ Such interpretability complements data-driven approaches by providing direct mechanistic insight alongside predictive capability.³⁴ Through this descriptor-driven framework, we provide general design principles for identifying efficient and selective NRR catalysts on layered materials and demonstrate a transferable strategy for accelerating catalyst discovery relative to exhaustive reaction pathway calculations.

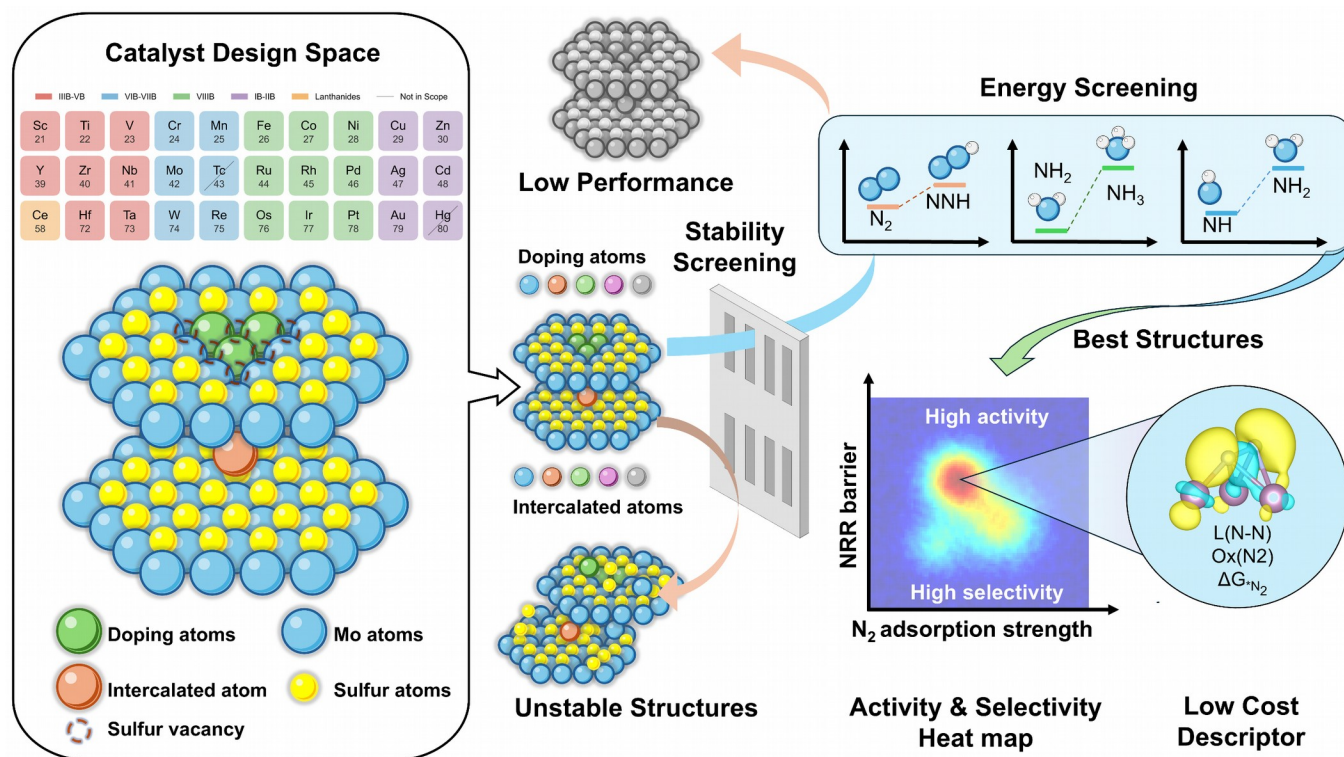


Figure 1. Schematic illustration of the catalyst design and multistep screening workflow for ID@MoS₂ (intercalated-and-doped MoS₂) catalysts for NRR. The design space is constructed by introducing dopant atoms (substituting Mo sites in the top layer) and intercalated atoms into a sulfur-vacancy-containing MoS₂ host, generating a combinatorial library of candidate structures. These candidates are first subjected to stability screening to eliminate unstable configurations and retain structurally viable ID@MoS₂ models. The stable structures are then evaluated through reaction-energy screening of key NRR intermediates (e.g., N₂, NNH, NH, NH₂ and NH₃) to identify energetically favorable pathways. The resulting candidates are mapped onto an activity–selectivity landscape (NRR barrier vs N₂ adsorption strength) to identify high-performance regions. Finally, the best structures are characterized using a low-cost descriptor set [e.g., L(N-N), charge of N₂, and ΔG_{*N_2}] to enable rapid prediction and interpretation of NRR activity/selectivity trends.

2. Methods

Density Functional Theory Calculations

Structure generation, workflow orchestration, and post-processing were performed using the CatGo platform³⁵. All density functional theory (DFT) calculations were performed using the Vienna Ab initio Simulation Package (VASP) with the projector augmented-wave (PAW) method.^{36–40} Exchange–correlation effects were treated within the generalized gradient approximation using the Perdew–Burke–Ernzerhof (PBE) functional.³⁶ To accurately capture the interlayer van der Waals interactions critical to the intercalated MoS₂ systems, the DFT-D3 dispersion correction of Grimme with Becke–Johnson damping (IVDW = 12) was applied throughout all calculations.⁴¹ A plane-wave kinetic energy cutoff of 450 eV was employed for all calculations.

The catalyst models were constructed using a bilayer MoS₂ supercell containing sulfur vacancies on the top layer. Substitutional doping was introduced by replacing three Mo atoms with a dopant element, while a single intercalated atom was placed in the interlayer region to form ID@MoS₂ structures. All geometries were fully optimized until the residual forces on each atom were below 0.05 eV Å⁻¹ and the total energy convergence criterion reached 10⁻⁵ eV. A vacuum spacing of at least 15 Å was applied perpendicular to the surface to avoid interactions between periodic images. Brillouin-zone sampling was performed using a 2 × 2 × 1 Monkhorst–Pack k-point mesh.⁴² Adsorption free energies were evaluated using the computational hydrogen electrode (CHE) model, including zero-point energy and entropic corrections for adsorbed intermediates.⁴³

Charge transfer between the catalyst surface and adsorbed N₂ was quantified using the Bader charge decomposition method as implemented by Henkelman and co-workers.^{44,45} The Bader charge on the adsorbed N₂ unit (relative to free N₂) was used to define the N₂ charge state, which serves as a key descriptor for electronic activation in the subsequent screening analysis. Chemical bonding was analyzed using the crystal orbital Hamilton population (COHP) method as implemented in the LOBSTER code.^{46,47} Wavefunctions from VASP calculations were projected onto localized atomic orbitals to compute COHP and integrated COHP (–ICOHP) values for N–N and metal–N interactions. The –ICOHP(N–N) values were used to quantify N≡N bond weakening, while –ICOHP(Metal–N) values measure metal–N₂ covalency. All bonding analyses were performed on fully relaxed adsorption geometries.

All free-energy diagrams use a constant-potential grand-canonical DFT (GC-DFT) framework. The electrode potential is set by adding or removing electrons ($N_e = -1.5, -1.0, -0.5, 0, +0.5, +1.0, +1.5 e^-$) compensated by a homogeneous background, with the aqueous environment treated by the VASPsol implicit solvent ($\epsilon_r = 80$, TAU = 0)^{48,49}. For every adsorbed intermediate we compute the grand potential $\Omega = G_{\text{DFT}} - E_F \cdot N_e$,^{50,51} where G_{DFT} is the SCF energy including zero-point and 298.15 K thermal/entropic corrections, and E_F the Fermi level. The electrode potential is obtained from the work function W_f as $U_{N_e}(\text{V/SHE}) = (W_f - 4.6 \text{ eV})/e$. The seven (U_{N_e}, Ω) points are fitted to $\Omega(U) = -\frac{1}{2} \times C(U - U_0)^2 + \Omega_0$, with C the effective interfacial capacitance (eV/V²) and U_0 the potential of zero charge. The pH dependence is introduced via the Nernst relation, $U(\text{V/RHE}) = U(\text{V/SHE}) + 0.0592 \cdot \text{pH}$ (pH = 7). Combined with the computational hydrogen electrode $\mu(\text{H}^+ + e^-) = \frac{1}{2} \times G(\text{H}_2) - eU_{\text{RHE}}$,⁵² the free-energy change of the k -th distal step is $\Delta G_k = \Omega(I_k, U) + n_{\text{NH}_3} \cdot G(\text{NH}_3) - \Omega(*\text{N}_2, U) - k \cdot [\frac{1}{2} \times G(\text{H}_2) - eU_{\text{RHE}}]$, with the gas-phase references $G(\text{N}_2)$, $G(\text{H}_2)$, $G(\text{NH}_3)$ held identical throughout. The onset potential is the least-negative U_{SHE} at which $\Delta G_k \leq 0$ for every distal step, located by binary search; the step that saturates this constraint is the potential-determining step (PDS).

Ab initio molecular dynamics (AIMD) simulations was performed with CP2K, using the PBE functional together with the DFT-D3(BJ) dispersion correction^{36,53–55}. The trajectories were propagated in the NVT ensemble at 298.15 K with a CSVR thermostat (time constant 80 fs), a 1.0 fs time step, and a total simulation time of 10 ps (10,000 steps). Brillouin-zone sampling was performed using a $1 \times 1 \times 1$ Monkhorst–Pack k -point mesh.⁴² The DZVP-MOLOPT-SR-GTH basis set and GTH-PBE pseudopotentials were used for all elements (S, Mo, Hf, W), with a cutoff energy of 450 Ry and a relative cutoff of 50 Ry.

Machine Learning and Descriptor Identification

To establish low-cost descriptors that capture NRR activity and selectivity trends without exhaustive pathway calculations, we employed a combination of machine learning regression and symbolic regression methods. Gradient-boosted tree models (XGBoost)⁵⁶ and extra-trees regressors⁵⁷ were trained on the DFT-computed dataset to predict key energetic quantities (e.g., limiting potential, selectivity metric) from a pool of candidate features derived from N₂ adsorption properties, electronic structure descriptors, and elemental attributes of the dopant and intercalant. Feature importance rankings from the ensemble tree models were used to identify the most informative descriptors. To obtain compact, physically interpretable analytic expressions, we further applied the Sure Independence Screening and Sparsifying Operator (SISSO) method,⁵⁸ which searches over algebraic combinations of primary features to yield low-dimensional descriptor formulas with quantified prediction errors. Model performance was assessed *via* leave-one-out cross-validation and parity analysis. Full details of hyperparameter settings, feature lists, and cross-validation results are provided in the **Supporting Information**.

3. Results and Discussion

3.1 Integrity of ID@MoS₂.

Prior to structural analysis, formation energies (E_f) were computed for all ID@MoS₂ candidates, with 94.3% of systems exhibiting negative E_f (**Figures S1 and S2**), confirming the overall thermodynamic accessibility of

the design space. Geometry optimization reveals that a substantial subset of ID@MoS₂ systems undergo structural reconstruction, indicating strong sensitivity of structural integrity to dopant and intercalant chemistry.^{59–61} Reconstruction patterns identified *via* K-means clustering and manual verification are summarized in **Tables S1-3 and Figure S3**. As shown in **Figure 2a**, dopants with smaller group numbers, such as Zr, Hf, Y, and W, consistently preserve structural integrity regardless of the intercalated species, whereas dopants with larger group numbers exhibit an increasing tendency toward reconstruction. The intercalated atom further modulates this behavior: V, Os, Fe, Co, and Ti generally maintain intact structures, while Ce, Pd, and Au frequently induce reconstruction, as summarized in **Figure S4a**. In addition, dopant identity strongly influences local geometric motifs, with smaller-group dopants forming equilateral triangular arrangements and larger-group dopants adopting distorted isosceles configurations, as shown in **Figure S4b-c**.

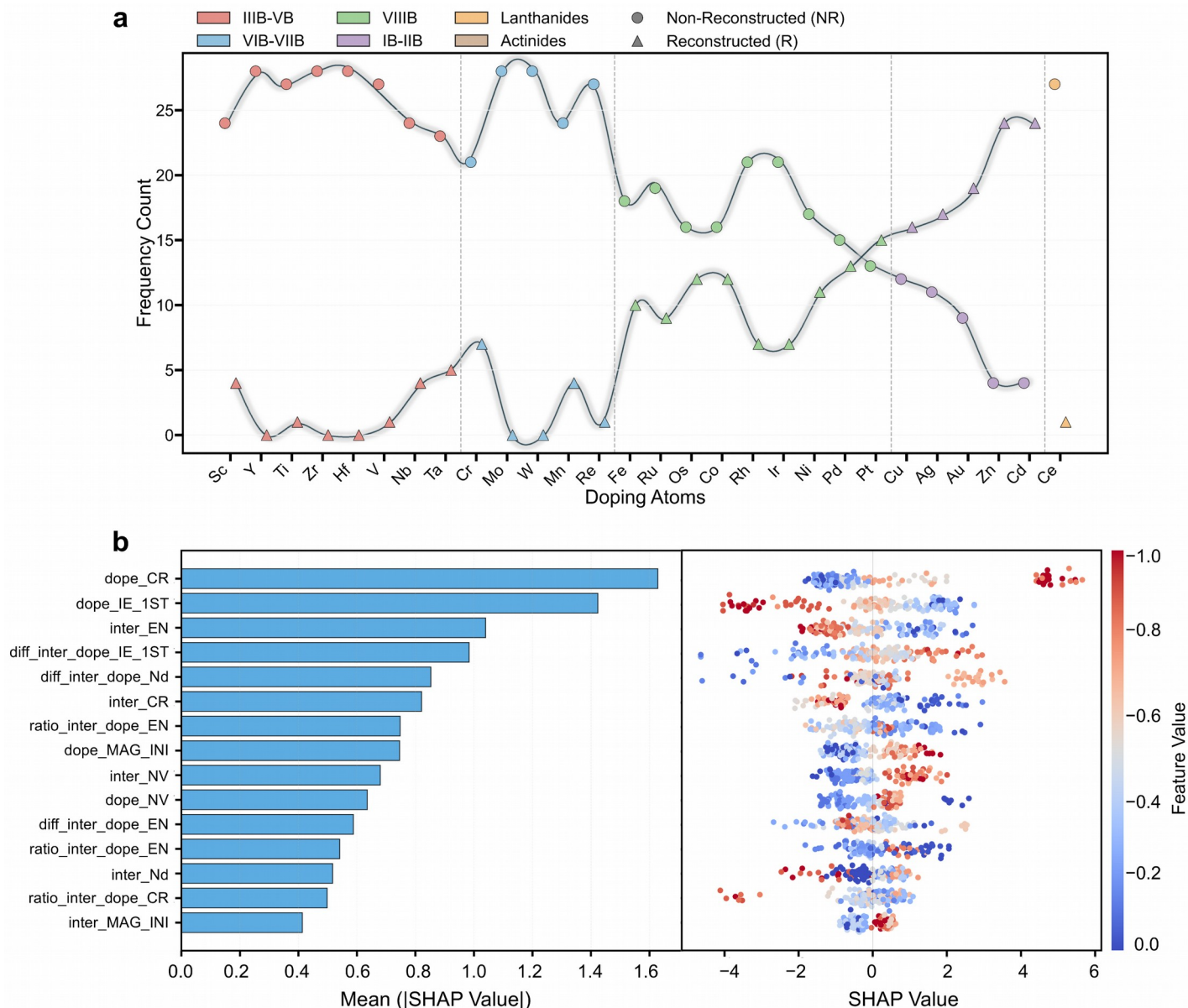


Figure 2. Structural integrity trends and descriptor analysis for ID@MoS₂ systems. (a) Dependence of structural integrity on dopant group and identity, showing the reconstruction behavior of ID@MoS₂ systems after geometry optimization. Symbols distinguish non-reconstructed and reconstructed structures, while shaded regions indicate different dopant group ranges. (b) Feature importance ranking derived from the Histogram-Based Gradient Boosting classifier and SHAP summary plot illustrates the contributions of individual atomic descriptors to the model prediction, where color denotes feature value and SHAP value reflects the impact on reconstruction probability.

Machine-learning classification based on intrinsic atomic descriptors confirms that these reconstruction behaviors are encoded in elemental properties rather than arising from stochastic relaxation pathways. Among

the tested models, the Histogram-Based Gradient Boosting classifier,^{62,63} exhibits the best performance, with evaluation metrics summarized in **Figure S5a-c** and classification results shown in **Figure S5d**. The model reliably distinguishes between intact and reconstructed structures, with an ROC AUC of 0.881. SHAP analysis⁶⁴ in **Figure 2b** identifies the dopant covalent radius and first ionization energy as the dominant factors governing structural integrity, followed by dopant electronegativity and the ionization energy mismatch between dopant and intercalant. The SHAP beeswarm plot further demonstrates that larger dopant size and higher ionization energy stabilize the structure, whereas highly electronegative intercalants and large electronic mismatches promote reconstruction.

Guided by these dominant features, SISO⁵⁸ yields a low-dimensional, physically interpretable descriptor that integrates intercalant electronegativity with dopant covalent radius and electronegativity, as shown in **Figure S6b**. While this descriptor captures the overall reconstruction trend and reproduces the observed dependence on dopant group number, a noticeable overlap between reconstructed and non-reconstructed structures remains, particularly within the nominally stable region shown in **Figure S6b-d**. This overlap indicates that reconstruction in ID@MoS₂ is governed by intrinsically non-linear, multi-factorial effects beyond the scope of a single low-dimensional descriptor. Nevertheless, the SISO expression consistently reinforces the central conclusion that increasing dopant electronegativity and decreasing dopant size systematically destabilize the structure, fully consistent with the trends observed in **Figure 2a** and the SHAP analysis in **Figure 2b**.

3.2 Activity and Selectivity Analysis

3.2.1 Effect of Intercalation on Activity, Selectivity, and Kinetics on Mo Sites

Guided by the structural integrity analysis in **Figure 2**, we focus on pristine Mo-doped MoS₂ surfaces and systematically vary the intercalated atom, as these systems remain structurally intact across all cases. This strategy enables a clean assessment of how intercalation alone modulates NRR activity, selectivity, and kinetics without complications from structural reconstruction.

Possible NRR reaction pathways on Mo-based sites are summarized schematically in **Figure 3a**. Although all four standard associative mechanisms (distal, alternating, enzymatic, and consecutive) were evaluated,⁶⁵ pathway calculations reveal that only the distal and enzymatic routes that yield the lowest limiting potentials and therefore govern the overall kinetics. A representative example is shown in **Figure 3b** for Sc-intercalated MoS₂, where the reaction proceeds preferentially via a sequential hydrogenation pathway. The calculated limiting potential for NRR is -0.17 V, while the competing HER exhibits a substantially more negative limiting potential of -0.72 V, yielding a selectivity window of 0.55 V. This pronounced separation demonstrates that appropriate intercalation can effectively suppress HER while maintaining high NRR activity.

To generalize these observations, the limiting potentials for NRR and HER are summarized for all intercalated systems in **Figure 3c**, which maps catalytic activity and selectivity in a unified framework. While most intercalated MoS₂ catalysts exhibit relatively favorable NRR activity, with U_L (NRR) spanning a narrow range of approximately 0.2 V, their selectivity varies significantly. Notably, all highly selective systems cluster within the IIIB–VIB intercalant groups, indicating that intercalation with early transition metals systematically enhances NRR selectivity. In contrast, intercalants with larger group numbers generally fail to achieve sufficient separation between NRR and HER energetics, despite maintaining reasonable activity. These results indicate that intercalation primarily tunes selectivity, with comparatively modest variation in intrinsic NRR activity.

Analysis of the PDSs reveals a clear trend. As shown in **Figure 3d**, 64% of all candidates are limited by the NH to NH₂ hydrogenation step, whereas only 36% are limited by the initial N₂ to NNH activation. Importantly, no system is found to be limited by the NH₂ to NH₃ step, which is despite frequently assumed to be rate-determining in prior NRR studies.^{26,66–71} Moreover, all highly selective catalysts in **Figure 3c** exhibit NH to NH₂ as the PDS. A clear correlation with intercalant chemistry emerges: intercalants from the IIIB to VB groups consistently exhibit NH to NH₂ as the PDS, whereas N₂ to NNH becomes competitive only for intercalants with larger group numbers. This behavior demonstrates that the electronic nature of the intercalated atom governs not only activity and selectivity, but also the identity of the kinetic bottleneck. This finding revises the commonly adopted view of regarding universal NRR bottlenecks^{66,69} and suggests new opportunities for breaking established scaling relations⁶⁸ through rational intercalation design.

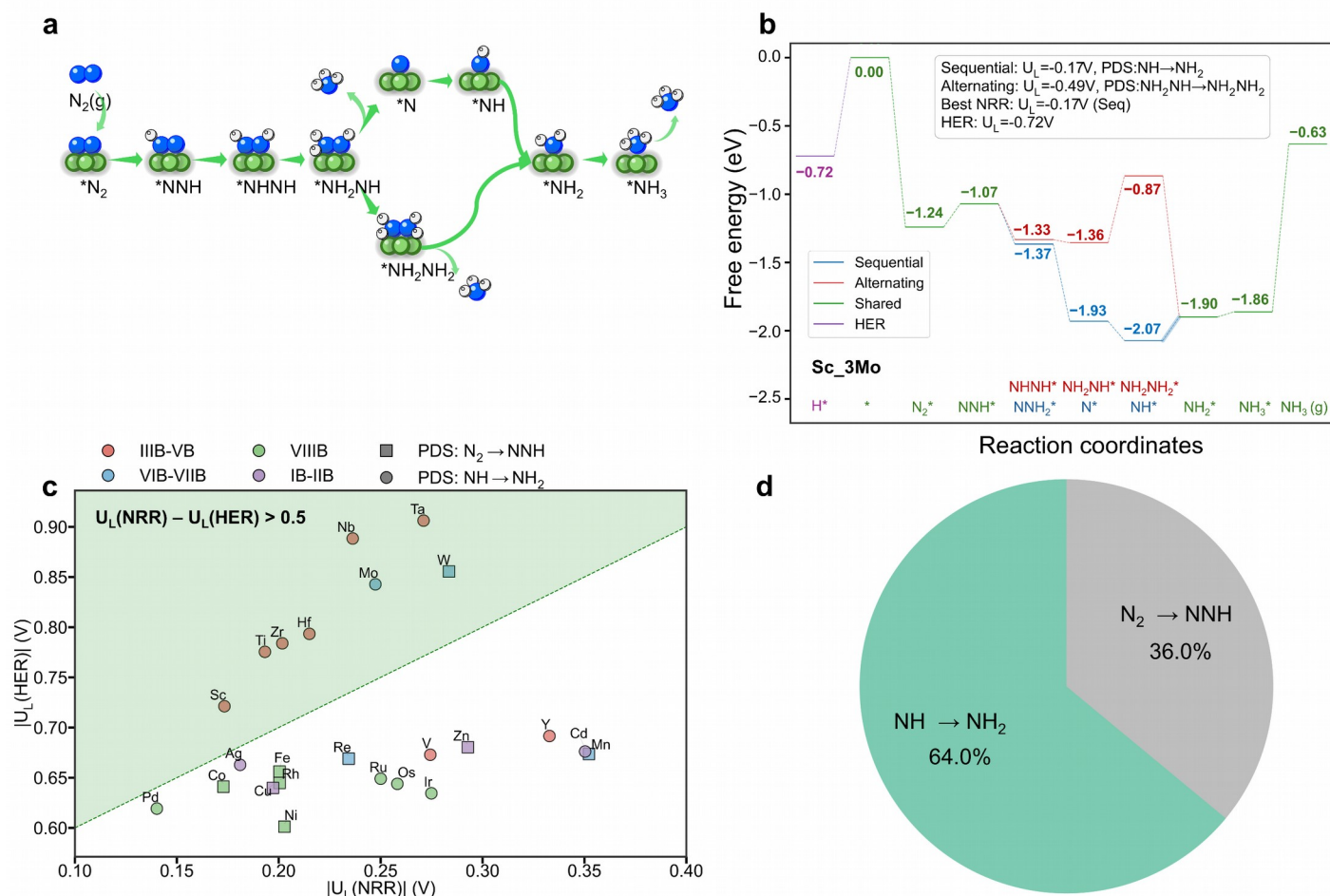


Figure 3. NRR pathways, energetics, and selectivity on intercalated MoS_2 with pristine Mo active sites. (a) Schematic illustration of possible associative nitrogen reduction pathways. (b) Free-energy diagram of NRR and HER on Sc-intercalated MoS_2 . (c) Activity-selectivity map of intercalated MoS_2 systems based on NRR and HER limiting potentials. (d) Statistical distribution of potential-determining steps for NRR across all intercalated systems.

3.2.2 NRR activities of all ID@ MoS_2

Motivated by the mechanistic insights obtained on pristine Mo-based surfaces, we extend the analysis to the full ID@ MoS_2 dataset by explicitly evaluating all key elementary steps governing NRR activity and selectivity. In most prior computational screening studies,^{66,67} only two elementary steps, namely $\text{N}_2 \rightarrow \text{NNH}$ and $\text{NH}_2 \rightarrow \text{NH}_3$, are considered as PDSs, as they define the two branches of the established NRR volcano.⁶⁶ However, our pathway analysis on pristine Mo-based surfaces reveals that the $\text{NH} \rightarrow \text{NH}_2$ hydrogenation step can also become the dominant kinetic bottleneck. We therefore evaluate the Gibbs free energies of all three candidate steps, $\text{N}_2 \rightarrow \text{NNH}$, $\text{NH} \rightarrow \text{NH}_2$, and $\text{NH}_2 \rightarrow \text{NH}_3$, together with the adsorption free energy of H^* to determine the HER limiting potential.

The resulting NRR activity landscape is summarized in **Figure 4a**. High NRR activity is predominantly observed for systems doped with Mo, W, Re, or Ti, indicating that the dopant largely determines the intrinsic activity of ID@ MoS_2 . These dopants are all early transition metals, consistent with their stronger binding to nitrogen-containing intermediates and favorable d-electron configurations for N_2 activation.^{65,72–75} In contrast, changing the intercalated atom modulates the activity only within a narrower range, indicating that intercalation serves as a secondary tuning parameter rather than the primary determinant of activity.

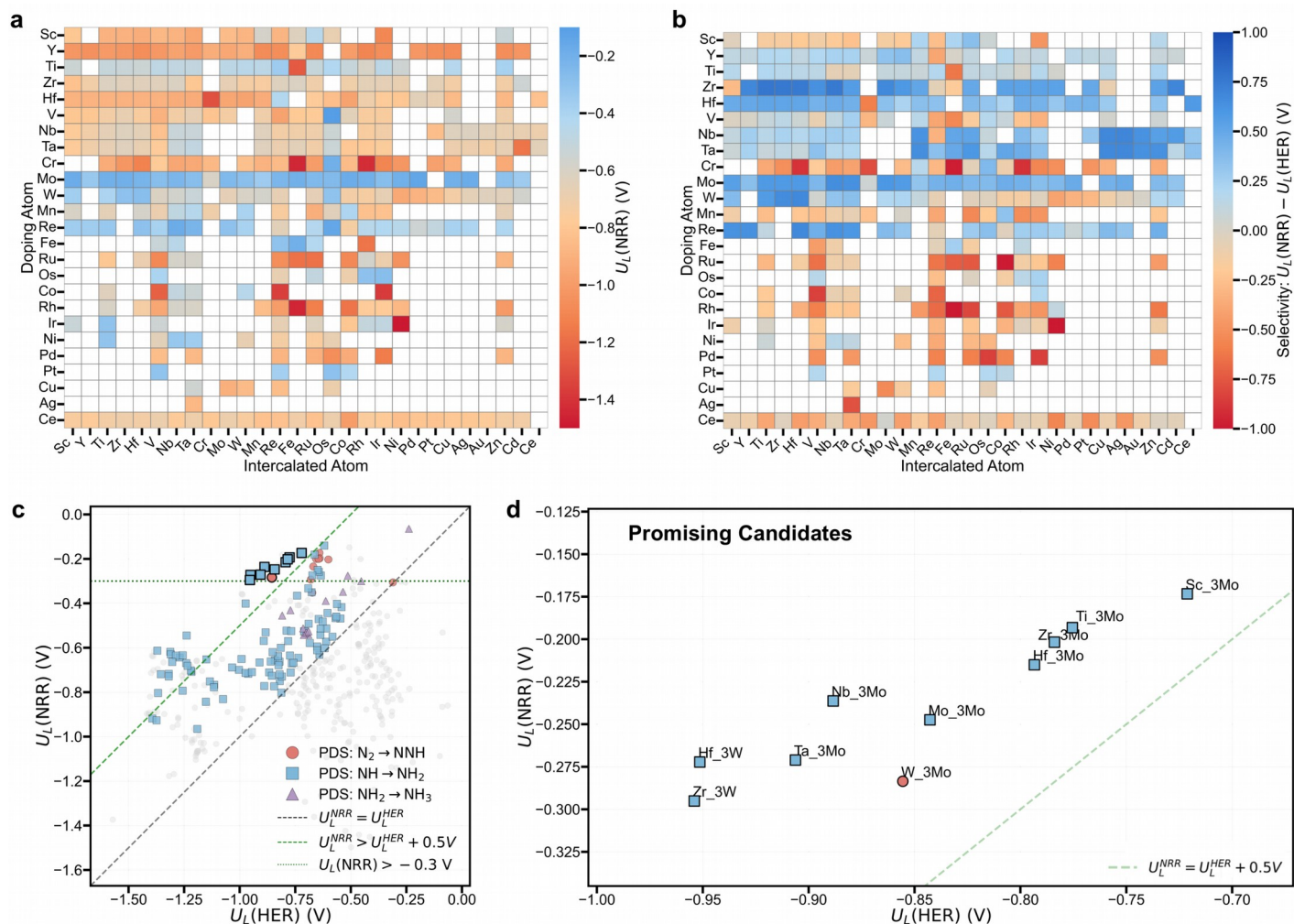


Figure 4. Activity and selectivity screening of the full ID@MoS₂ dataset. (a) Heat map of NRR limiting potentials for all ID@MoS₂ systems. (b) Heat map of NRR selectivity quantified by the difference between NRR and HER limiting potentials. (c) Distribution of NRR activity and selectivity for all systems, with symbols indicating the potential-determining step. (d) Activity–selectivity map of the screened promising catalysts.

Figure 4b illustrates how the doped and intercalated atoms cooperatively regulate NRR selectivity. Highly selective systems are predominantly associated with early transition-metal dopants (i.e., elements located before Mn in the periodic table), indicating that smaller dopant group numbers systematically favor NRR selectivity over HER. In contrast, dopants with larger group numbers generally exhibit poor selectivity, even when their NRR activity remains moderate. The absence of data points in certain regions reflects structural reconstruction occurring either for the bare surface or for key intermediates during optimization, highlighting that early transition-metal dopants also provide superior structural robustness under reaction conditions. The influence of intercalation on selectivity is further analyzed in **Figure S7**, where group VIII intercalants tend to yield inferior selectivity, while most other intercalant groups show comparable selectivity. Notable exceptions include Cr and Re intercalation, which also result in reduced selectivity.

The combined activity and selectivity distribution for all systems is shown in **Figure 4c**. Gray data points correspond to candidates that can already be excluded based on previously established screening criteria (see Supporting Information). Building on this baseline, we impose more stringent filters by requiring an activity threshold of $U_L(\text{NRR}) \geq -0.30$ V and a selectivity criterion of $U_L(\text{NRR}) - U_L(\text{HER}) \geq 0.50$ V. The activity threshold is consistent with the best limiting potentials reported for state-of-the-art NRR electrocatalysts in recent computational studies,^{66,76,77} while the selectivity window exceeds the minimum difference typically required to suppress the competing HER under electrochemical conditions.^{65,78–80} These stricter criteria narrow the candidate pool to a smaller subset of catalysts that simultaneously achieve high activity and high selectivity.

Analysis of the PDS for the screened systems reveals a clear mechanistic trend. As evident from **Figure 4c**, the majority of candidates passing the conventional screening criteria are limited by the $\text{NH} \rightarrow \text{NH}_2$ step, while only a minority are limited by $\text{N}_2 \rightarrow \text{NNH}$ or $\text{NH}_2 \rightarrow \text{NH}_3$. Notably, systems for which $\text{NH}_2 \rightarrow \text{NH}_3$ is the PDS consistently exhibit poor selectivity, in agreement with the trends observed on pristine Mo surfaces. This consistency indicates that $\text{NH} \rightarrow \text{NH}_2$ constitutes a general kinetic bottleneck across structurally intact ID@MoS₂ systems.

The final set of ten promising catalysts identified under the refined screening criteria is shown in **Figure 4d**. Remarkably, all selected candidates contain either Mo or W as the doped atom, while the intercalated atoms are exclusively drawn from the IIB–VIB early transition-metal groups. This convergence strongly suggests that both NRR activity and selectivity in ID@MoS₂ are governed by group-dependent chemical properties of the doped and intercalated atoms. Together, these results establish a clear design principle in which early transition-metal dopants define the intrinsic activity, while intercalation with chemically compatible early transition metals further optimizes selectivity and kinetic favorability.

To further confirm the dynamic stability of these candidates under operating conditions, representative systems (Hf@Mo/MoS₂ and Hf@W/MoS₂) were subjected to AIMD simulations at 298.15 K for 10 ps (**Figure S8**). Both systems reach a sub-Ångström root-mean-square displacement (RMSD) plateau (~ 0.15 Å, all-atom) within 0.2 ps and maintain intact active-site architecture throughout the trajectory, confirming their thermal stability under reaction conditions.

3.3 Descriptor Analysis for Rapid Screening of Active and Selective NRR Catalysts

Building on the activity and selectivity screening in **Figure 4**, we next assess whether low-dimensional energetic descriptors can accelerate the identification of promising NRR catalysts without exhaustive pathway calculations.

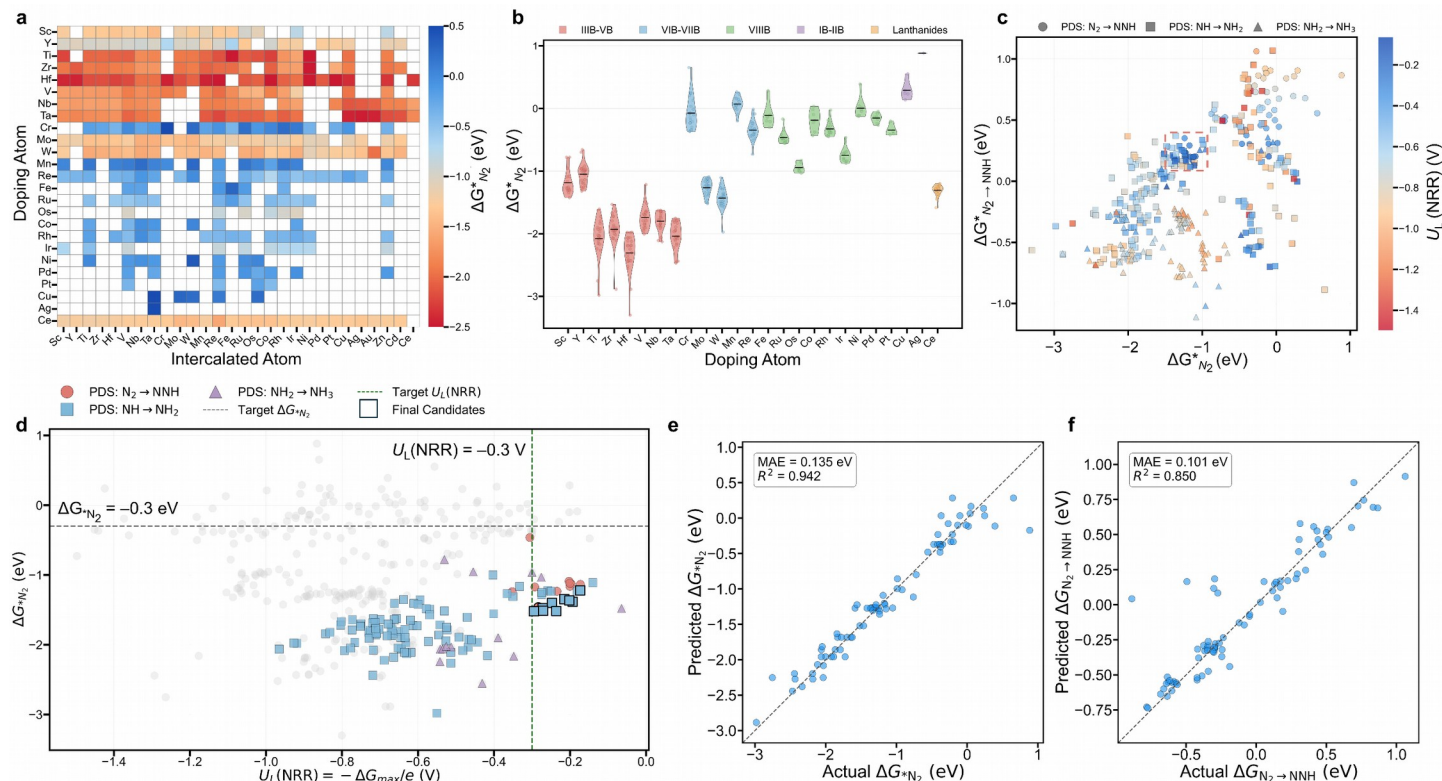


Figure 5. Energetic descriptor analysis for NRR screening. (a) Heat map of N₂ adsorption free energies for all ID@MoS₂ systems. (b) Statistical distribution of N₂ adsorption free energies grouped by dopant family. (c) Two-dimensional descriptor map based on N₂ adsorption and protonation free energies. (d) N₂ adsorption free energies of the screened promising catalysts. (e) Predicted versus calculated N₂ adsorption free energies from machine learning models. (f) Predicted versus calculated N₂ protonation free energies from machine learning models.

One-dimensional descriptor: N₂ adsorption free energy. The adsorption free energy of N₂ is widely recognized as a key descriptor for nitrogen activation.^{81,66,82,65} As shown in **Figure 5a**, N₂ adsorption across the full ID@MoS₂ dataset depends strongly on the group number of the dopant: IVB–VIB dopants bind N₂ most strongly, IIIB dopants more weakly, and dopants beyond group VIB substantially less favorably. Within each group, adsorption strengthens with increasing period number (**Figure 5b**), tracking the larger covalent radii of heavier elements. In contrast, the intercalant exhibits no clear monotonic trend with either group or period (**Figure 5a**), confirming that the dopant, not the intercalant, primarily governs N₂ activation strength. However, while all promising catalysts identified in **Figure 4d** cluster within a narrow N₂ adsorption window of −1.0 to −1.6 eV (**Figure 5d**), many non-promising candidates also fall in this range. Thus, N₂ adsorption alone is insufficient to identify catalysts that simultaneously achieve high activity and selectivity.⁷⁷

Two-dimensional descriptors: activity and selectivity. To improve resolution, we pair N₂ adsorption free energy with the free energy of each candidate potential-determining step (**Figure 5c**). Combining N₂ adsorption with the first protonation step (N₂ → N₂H) provides the most effective activity criterion: promising systems cluster tightly within a window where $\Delta G(\text{N}_2 \rightarrow \text{N}_2\text{H})$ lies between ~0.1 and 0.4 eV, cleanly excluding most low-activity candidates. The NH → NH₂ pairing fails to separate active and inactive systems, while the NH₂ → NH₃ pairing yields a comparable activity window but requires an additional adsorbed intermediate, increasing computational cost without compensating gain. In contrast, no two-dimensional combination of N₂ adsorption with a single protonation step reliably distinguishes selective from non-selective systems (**Figure S9**). Poorly selective candidates invariably overlap with selective ones in every tested descriptor pair, reflecting the inherently coupled competition between NRR and HER, which cannot be captured by NRR-side energetics alone.

Machine-learning pre-screening. Finally, we test whether these key descriptors can be predicted directly from intrinsic atomic features, bypassing DFT entirely for first-stage screening. Decision-tree and random-forest models predict $\Delta G(^*\text{N}_2)$ and $\Delta G(\text{N}_2 \rightarrow \text{N}_2\text{H})$ with mean absolute errors of 0.135 and 0.101 eV, respectively (**Figures 5e** and **5f**). SHAP analysis (**Figure S10**) confirms that both predictions are dominated by dopant-related features: dopant covalent radius most strongly controls N₂ adsorption strength (larger radii → stronger adsorption), while dopant group number governs protonation energetics (smaller groups → lower barriers). These trends are consistent with the chemical analysis above and validate machine-learning descriptors as an efficient first-stage screening tool.

3.4 Electronic Structure Analysis

To elucidate how intercalation modulates NRR activity and selectivity, we analyze the electronic structure of intercalated pristine MoS₂ systems, where structural reconstruction is absent and purely electronic effects can be isolated. The Mo–N₂ interaction is governed by two complementary bonding channels (**Figure 6**): π -back-donation from occupied Mo d states into N₂ π^* orbitals, which weakens the N≡N bond and drives activation, and σ -donation from the N₂ lone pair into unoccupied Mo d orbitals, which determines adsorption strength. We show below that intercalation tunes these two channels independently, and that their interplay rationalizes the periodic trends observed across the intercalant series.

Decoupling N₂ activation from intercalant-to-host charge transfer. **Figure 6a-c** track N–N bond elongation against three Bader charge quantities. The total charge transferred from the intercalant to the MoS₂ host, $\Delta q(\text{M})$, correlates poorly with bond elongation (**Figure 6a**): early transition-metal intercalants (IIIB–VB) donate substantial charge yet yield relatively short N–N bonds, whereas late transition-metal intercalants (VIIB) donate less but induce greater elongation. The Mo oxidation state $\Delta q(\text{Mo})$ correlates positively with N–N bond length ($R^2 = 0.85$, **Figure 6b**), counterintuitively, activation increases as the Mo center becomes less reduced. The strongest correlation is with charge accumulated on N₂ itself, $\Delta q(\text{N}_2)$ ($R^2 = 0.98$, **Figure 6c**). Together, these results show that N₂ activation depends not on the total electron density at Mo, but on the ability of Mo d states to efficiently channel charge into N₂ π^* orbitals via back-donation.

Two complementary descriptors capture activation and adsorption. In contrast to activation, the N₂ adsorption free energy correlates only moderately with $\Delta q(\text{N}_2)$ ($R^2 = 0.42$, **Figure 6d**), indicating that charge transfer alone is insufficient to capture binding strength. This is expected from the dual bonding picture above:

adsorption depends on both back-donation (controlled by occupied Mo d-state energetics) and σ -donation (controlled by the availability of unoccupied Mo d states), as shown in **Figure 6e**. We therefore introduce two electronic descriptors that target each channel separately: the d-band center of the hybridized Mo–M states, ϵ_d (**Figure 6f**), governs π -back-donation efficiency, while the integrated unoccupied d-state density within the [0, 2] eV window above the Fermi level, n_d^{unocc} (**Figure 6g**), governs σ -donation capacity. Across the intercalant series, these two quantities follow opposing periodic trends, producing a systematic trade-off between activation and adsorption.

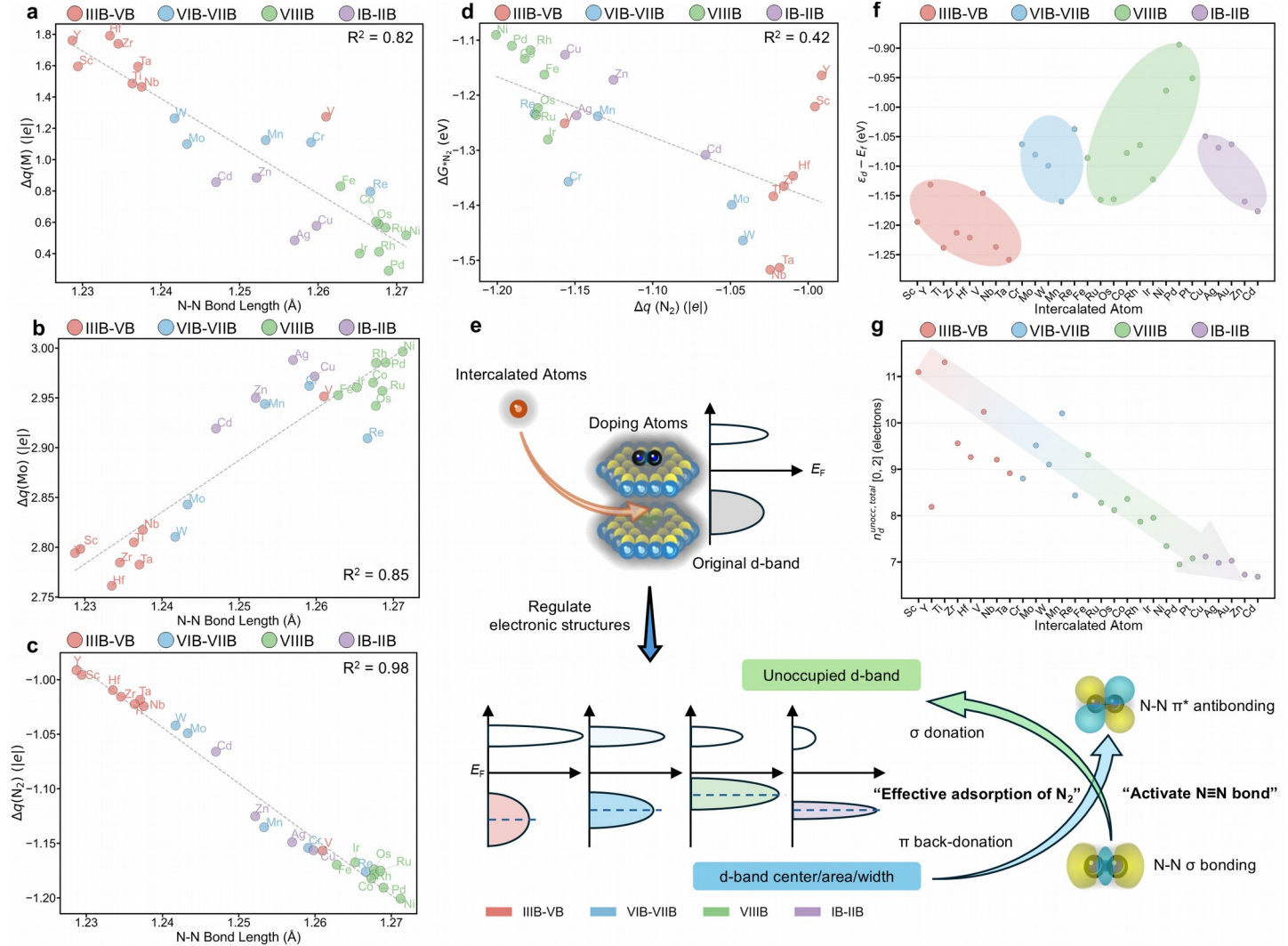


Figure 6. Electronic structure descriptors for N_2 adsorption on intercalated pristine MoS_2 systems. (a–c) Correlations between the N–N bond length of adsorbed N_2 and the Bader charge difference (Δq) of (a) the intercalated atom M, (b) the catalytically active Mo site, and (c) the adsorbed N_2 molecule, respectively. (d) Correlation between the N_2 adsorption free energy ($\Delta G(*N_2)$) and the charge transferred to $*N_2$. (e) Schematic illustration of how intercalation regulates the electronic structure of the Mo active site, modulating the relative contributions of σ -donation from the N_2 lone pair into unoccupied Mo d-states and π -back-donation from occupied Mo d-states into N_2 π orbitals. (f) Projected d-band center (ϵ_d) of the hybridized Mo–M states at the active site for each intercalant. (g) Integrated unoccupied d-electron density of the hybridized Mo–M states within the [0, 2] eV window above the Fermi level. In all panels, data points are colored by the periodic group of the intercalation atom: IIIB–VB (red), VIB–VIIB (blue), VIIIB (green), and IB–IIB (purple). Dashed lines in (a–d) denote linear fits, with R^2 values indicated.

Periodic trends across intercalant groups. The opposing behavior of ϵ_d and n_d^{unocc} rationalizes the four observed regimes, as shown in **Figure 6e**:

- 1) For early transition-metal intercalants (IIIB–VB), the ϵ_d remains relatively low and is not substantially shifted upward by hybridization, resulting in limited π -back-donation and weaker N_2 activation. However, these elements possess abundant n_d^{unocc} in the [0, 2] eV window, which enhance σ -donation

upon hybridization and yield strong N₂ adsorption. This explains why systems such as Sc- and Y-intercalated MoS₂ exhibit strong adsorption but limited N–N bond activation.

- 2) Moving to intermediate intercalants (VIB–VIIB), moderate d-electron occupation in the [−2, 0] eV region produces modest upward shifts in the composite ε_d , enhancing back-donation without eliminating unoccupied d states. At the same time, these intercalants retain a moderate density of n_d^{unocc} in the [0, 2] eV window, providing a moderate enhancement of σ -donation capacity. The resulting balance between π -back-donation and σ -donation leads to concurrent N₂ activation and adequate adsorption, consistent with the favorable activity–selectivity profiles identified in the screening results (**Figures 3-5**).
- 3) For late transition-metal intercalants (VIIB), dense occupied d states near the Fermi level shift ε_d upward more significantly, strengthening π -back-donation and promoting pronounced N–N bond activation. However, these intercalants possess few unoccupied d states (n_d^{unocc}) above the Fermi level, limiting σ -donation and weakening adsorption. Consequently, late transition-metal intercalants activate N₂ effectively but bind it relatively weakly.
- 4) Finally, IB–IIB intercalants represent a further evolution of this trend. Their d states are highly localized and lie deeper in energy; hybridization sharpens the density of states near the Fermi level, enabling moderate back-donation. However, the scarcity of unoccupied d states constrains σ -donation, resulting in comparatively weak adsorption.

Collectively, the interplay between ε_d and n_d^{unocc} provides a unified electronic framework: early intercalants favor adsorption through enhanced σ -donation, late intercalants favor activation through elevated π -back-donation, and optimal NRR performance arises from intercalants that balance these competing channels. This framework explains why the IIB–VIB intercalation window is electronically optimal, closing the loop between the energetic descriptors and the underlying orbital chemistry.

3.5 Generalization of N₂-Based Descriptors across the Full ID@MoS₂ Dataset

Motivated by the descriptor trends established for pristine Mo-based systems, we next examine whether these N₂-centered descriptors can be generalized to the full ID@MoS₂ systems. Specifically, we evaluate the combined behavior of the N₂ charge state ($\Delta q(\text{N}_2)$), N–N bond length, and N₂ adsorption free energy ($\Delta G^*_{\text{N}_2}$) as a unified screening framework across systems with different doped and intercalated atoms.

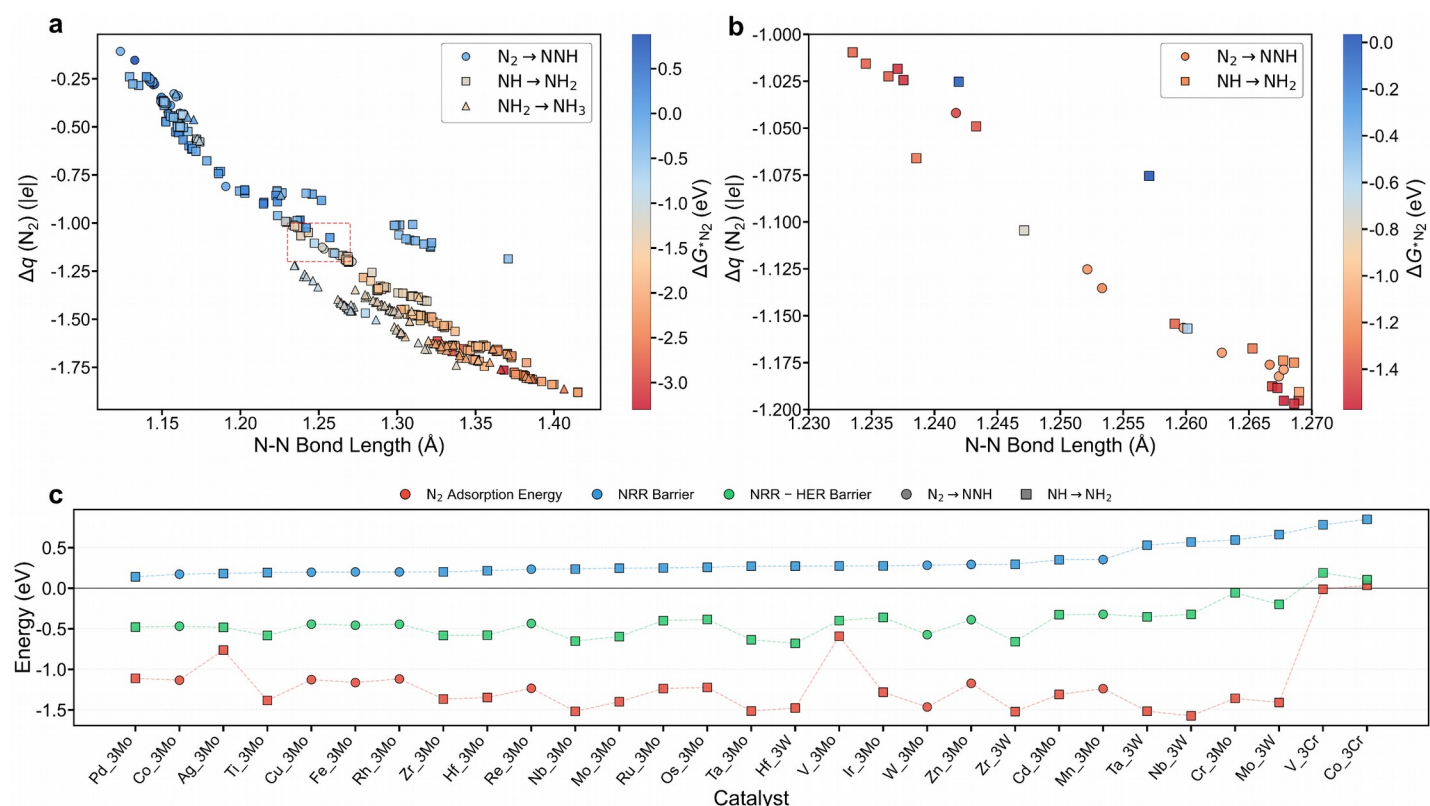


Figure 7. Generalization of N₂-based descriptors across the full ID@MoS₂ dataset. (a) Correlation between N₂ charge state Δq (N₂), N–N bond length, and N₂ adsorption free energy $\Delta G^{\circ}_{N_2}$ for all systems. (b) Selected descriptor window based on N–N bond length and Δq (N₂). (c) Relationship between N₂ adsorption free energy, NRR activity, and NRR–HER selectivity for the screened systems.

As shown in **Figure 7a**, a clear correlation emerges across all systems: decreasing Δq (N₂) is accompanied by systematic N–N bond elongation and stronger N₂ adsorption. This trend indicates that charge transfer to N₂, structural activation, and adsorption energetics remain intrinsically coupled even across chemically diverse ID@MoS₂ configurations. Importantly, this relationship holds regardless of dopant or intercalant identity, demonstrating that these N₂-centered descriptors capture a common physical response to electronic perturbation of the active site. Guided by the optimal activation window identified for pristine Mo systems (**Figure 7a**), we focus on a restricted region defined by N–N bond lengths of approximately 1.23–1.27 Å and N₂ charge states between -1.0 and -1.2 (**Figure 7b**). This window corresponds to moderate yet effective N₂ activation, avoiding both insufficient activation and excessive charge localization. Notably, most catalysts falling within this window are well consistent with those previously identified as promising based on explicit activity and selectivity calculations.

Further analysis of supports the effectiveness of this descriptor space. As shown in **Figure 7c**, systems with N₂ adsorption free energies stronger than -1.0 eV consistently exhibit favorable NRR activity, while maintaining a NRR–HER limiting-potential difference of ~ -0.5 eV, indicative of strong selectivity. This convergence demonstrates that catalysts selected using the combined N₂ charge state, N–N bond length, and adsorption free energy criteria achieve both high activity and effective suppression of competing HER.

Collectively, this three-dimensional descriptor provides a robust and transferable screening framework for identifying high-performance NRR catalysts across the ID@MoS₂ chemical space. Unlike one-dimensional energetic descriptors, this combined descriptor captures complementary aspects of adsorption strength, electronic activation, and structural response of N₂, enabling efficient pre-screening without explicit evaluation of full reaction pathways.

3.6 Validation of Top Candidates via Grand Canonical DFT

The descriptor-driven screening presented above identifies Hf@Mo/MoS₂ (Hf-intercalated MoS₂ retaining the native Mo active center, stoichiometry Hf₁Mo₃₂S₅₈) and Hf@W/MoS₂ (Hf-intercalated, W-substituted MoS₂, stoichiometry Hf₁Mo₂₉W₃S₅₈) as the two most promising ID@MoS₂ candidates for NRR. To validate these predictions under more realistic electrochemical conditions, we carried out two complementary investigations: (i) AIMD simulations to assess thermal stability and (ii) grand canonical DFT (GC-DFT) calculations to map the full potential-dependent NRR free-energy landscape.

3.6.1 Thermal Stability Assessment

A prerequisite for any practical NRR catalyst is that the as-designed active-site architecture remain structurally intact under operating conditions. We therefore subjected both top descriptor-screening candidates to AIMD simulations under realistic finite-temperature electrochemical conditions.

For Hf@Mo/MoS₂, the AIMD trajectory shows that the intercalated Hf atom remains firmly coordinated within the van der Waals gap throughout the entire 10 ps run, with no tendency toward surface migration or interlayer ejection. The total-energy fluctuations stay within ± 3 meV/atom over 9 ps of production, and the temperature oscillates stably about its 298 K set point (**Figure 8a-c**). For Hf@W/MoS₂, the three substitutional W atoms retain their lattice positions throughout the trajectory and the Hf intercalant likewise remains structurally anchored (**Figure 8d-f**). The atomic RMSD converges within the first ~ 0.2 ps in both systems, confirming rapid thermalization and preserved structural integrity (**Figure S8**). These results establish that the Hf-intercalated configurations identified by the screening are thermally robust under ambient electrochemical conditions.

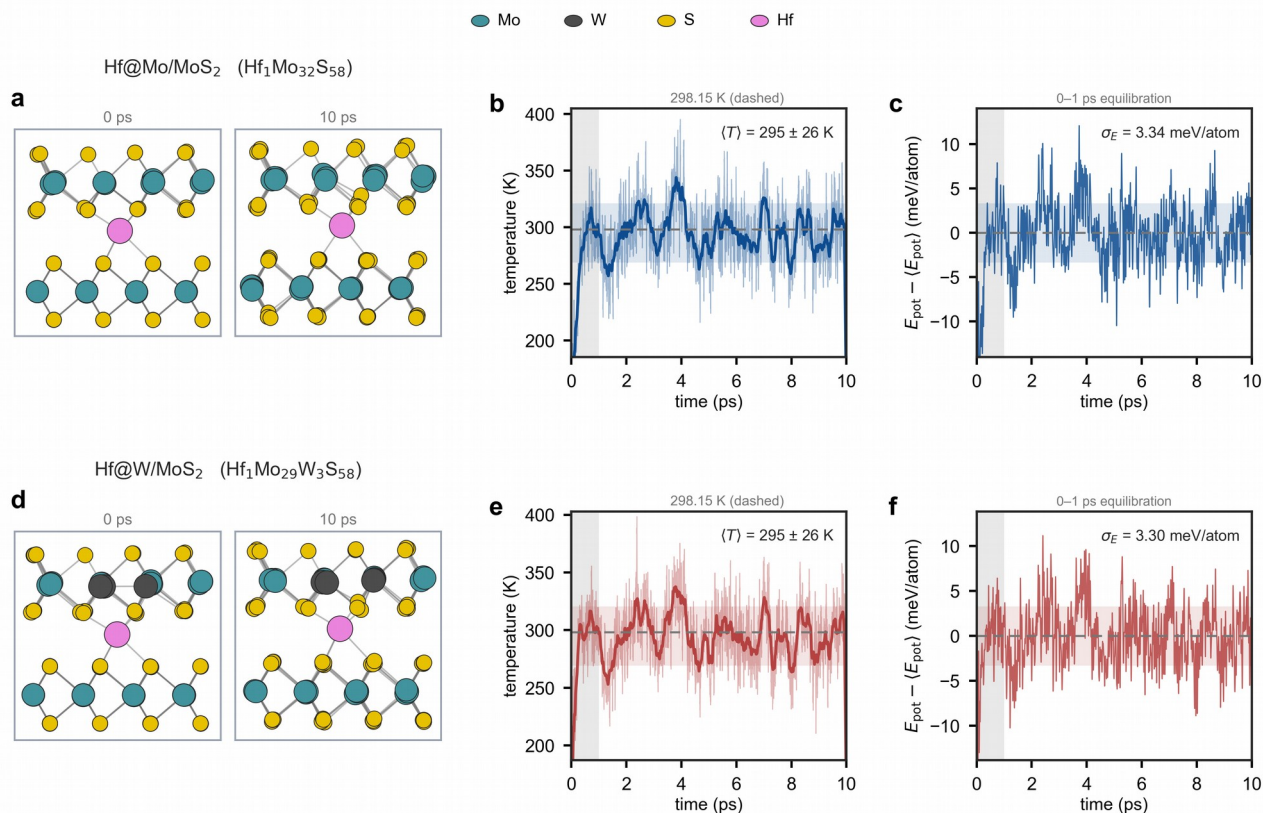


Figure 8. Thermal stability of the two top NRR candidates from 10 ps ab initio molecular dynamics. (a) Hf@Mo/MoS₂ side-view structure at 0 ps vs 10 ps; (b) instantaneous temperature and (c) per-atom potential-energy fluctuation ($E_{\text{pot}} - \langle E_{\text{pot}} \rangle$) versus time for Hf@Mo/MoS₂. (d–f) The corresponding panels for Hf@W/MoS₂. In the structure panels, Mo (teal), W (dark grey), S (yellow) and the intercalated Hf (magenta) retain their positions, with the Hf atom remaining anchored in the van der Waals gap; thin sticks denote metal–S bonds. Solid lines in (b,c,e,f) are 0.15 ps running means over the raw trace (faint); the shaded band is $\pm 1\sigma$ of the 1–10 ps production window, the dashed line marks the 298.15 K set point (b,e) / the production mean (c,f), and the grey region marks the 0–1 ps equilibration that is excluded from all reported statistics.

3.6.2 Grand Canonical DFT Analysis of NRR Energetics

Hf@Mo/MoS₂. The seven distal-pathway intermediates (*N₂, *NNH, *NNH₂, *N + NH₃, *NH, *NH₂, *NH₃) were calculated at all seven charge states, yielding 49 single-point GC-DFT calculations (**Table S5**). The parabolic fits reproduce the data with excellent fidelity ($R^2 = 0.959\text{--}0.990$; $C = 1.23\text{--}1.77$ eV/V²; **Figure S11a**). The onset potential, which is defined as the least-overpotential U_{SHE} at which every elementary step satisfies $\Delta G \leq 0$, was located by binary search at -0.167 V vs RHE. The PDS is the first protonation. At the onset potential the cumulative free-energy release across the six-electron distal pathway is -1.90 eV, the post-PDS steps releasing -0.63 , -0.77 , -0.34 , -0.04 and -0.12 eV for *NNH $\rightarrow$ *NNH₂, *NNH₂ $\rightarrow$ *N + NH₃, *N $\rightarrow$ *NH, *NH $\rightarrow$ *NH₂ and *NH₂ $\rightarrow$ *NH₃, respectively (**Figure 9a**).

Hf@W/MoS₂. The same seven distal-pathway intermediates were evaluated analogously, giving 49 single-point GC-DFT calculations (**Table S5**). The parabolic fits again yield high R^2 ($0.952\text{--}0.986$; $C = 1.23\text{--}1.99$ eV/V²; **Figure S9b**). The onset potential is -0.624 V vs RHE, markedly more negative (by ≈ 0.46 V) than that of Hf@Mo/MoS₂, and the PDS shifts to the final hydrogenation step, *NH₂ $\rightarrow$ *NH₃. At the onset potential the cumulative free-energy release for the distal pathway is -4.13 eV, with step-wise ΔG of -0.66 , -0.97 , -1.30 , -0.72 , -0.48 and 0.00 eV for *N₂ $\rightarrow$ *NNH, *NNH $\rightarrow$ *NNH₂, *NNH₂ $\rightarrow$ *N + NH₃, *N $\rightarrow$ *NH, *NH $\rightarrow$ *NH₂ and *NH₂ $\rightarrow$ *NH₃, respectively (**Figure 9b**).

High-throughput descriptor screening followed by GC-DFT validation singles out Hf@Mo/MoS₂ as the immediate experimental target. Among the two finalists distilled from the full ID@MoS₂ library, Hf@Mo/MoS₂ exhibits the less negative onset potential (-0.167 V vs RHE) with *N₂ $\rightarrow$ *NNH as the PDS, a value lying

within the optimum window of Mo-based single-atom NRR catalysts identified in earlier high-throughput DFT screenings (typically -0.10 to -0.30 V vs RHE)^{66,72}. The first-protonation PDS and the balanced binding of all NRR intermediates confirm that the descriptor pipeline retains a catalyst whose thermodynamics are competitive with the published optimum, validating the two-stage descriptor and GC-DFT workflow for the accelerated discovery of intercalant–dopant MoS₂ NRR catalysts. Hf@W/MoS₂ is identified as a secondary candidate: although W activates N₂ exceptionally strongly, GC-DFT predicts an onset of -0.624 V vs RHE that is outside the experimentally accessible window of reported W-based single-atom NRR catalysts⁷². The bottleneck shifts to $*\text{NH}_2 \rightarrow *\text{NH}_3$ because the strong d–N interaction at W overbinds late N_xH_y intermediates, making product release rate-limiting, which is a known limitation of late-transition-metal SACs that future electronic tuning could alleviate⁷². Collectively, these GC-DFT results validate Hf@Mo/MoS₂ as the priority target for experimental verification and relegate Hf@W/MoS₂ to a backup motif for further site engineering.

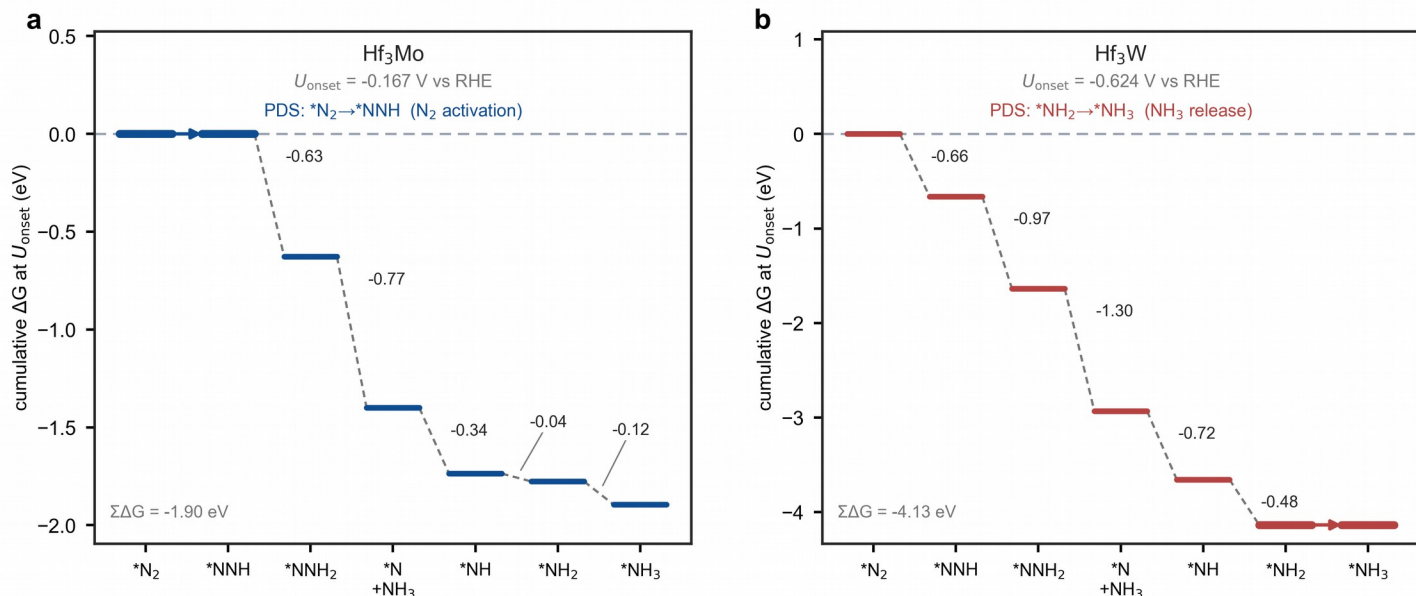


Figure 9. Grand canonical free-energy diagram for the distal NRR pathway at each catalyst's onset potential. (a) Hf@Mo/MoS₂ at $U_{\text{onset}} = -0.167$ V vs RHE: the first protonation $*\text{N}_2 \rightarrow *\text{NNH}$ is the potential-determining step. (b) Hf@W/MoS₂ at $U_{\text{onset}} = -0.624$ V vs RHE: the bottleneck shifts to the final protonation $*\text{NH}_2 \rightarrow *\text{NH}_3$. Horizontal bars are the cumulative ΔG of each intermediate; dashed connectors are labelled with the step-wise ΔG (eV); $\Sigma\Delta G$ is the total driving force at onset.

4. Conclusion

This work establishes a unified mechanistic and descriptor-based framework for nitrogen reduction on intercalated and doped MoS₂. Through reconstruction-aware screening of 784 dopant–intercalant combinations, combined with electronic-structure analysis and interpretable machine learning, we resolve a clear division of labor between the two engineering strategies: the dopant dictates intrinsic NRR activity, with Mo and W identified as uniquely effective active centers, while intercalation acts as a remote electronic regulator that modulates selectivity and kinetics from within the van der Waals gap. Mechanistic analysis identifies $\text{NH} \rightarrow \text{NH}_2$ as the dominant potential-determining step across most active and selective systems, revising the long-standing emphasis on early ($\text{N}_2 \rightarrow \text{NNH}$) or late ($\text{NH}_2 \rightarrow \text{NH}_3$) hydrogenation steps and highlighting an overlooked kinetic bottleneck in NRR.

We further show that N₂ adsorption free energy alone is insufficient for screening, and instead establish a transferable three-dimensional descriptor space: N₂ adsorption free energy, N₂ charge state, and N–N bond length, that robustly captures both activity and selectivity across the full chemical space. Electronic-structure analysis traces these trends to intercalation-controlled redistribution of Mo-projected d-states near the Fermi level, which tunes the balance between σ -donation and π -back-donation at the active site and gives rise to a non-monotonic dependence of N₂ activation on the d-band characteristics of the Mo center. Grand-canonical DFT

validation under realistic electrochemical conditions singles out Hf-intercalated Mo-MoS₂ as a priority experimental target, with an onset potential of −0.17 V vs RHE.

Together, these results provide generalizable design principles for NRR on layered catalysts via cooperative doping and intercalation, and demonstrate an efficient descriptor-driven workflow for accelerating catalyst discovery beyond exhaustive reaction pathway calculations. The same framework, distinguishing the surface-localized role of dopants from the remote electronic role of intercalants, should be broadly applicable to other layered-material electrocatalysts where activity and selectivity must be tuned independently.

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